Session 1: Solutions

Lesson exercises. Do not open until your Exercise 4 write-up is done.

Exercises on the lesson

Exercise 1. Write $n = ab$ with $1 < a, b < n$. By identity (a) with a^b in place of a and 1 in place of b: $2^n - 1 = (2^a)^b - 1 = (2^a - 1)\left((2^a)^{b-1} + \cdots + 1\right)$. The first factor satisfies $1 < 2^a - 1 < 2^n - 1$, so $2^n - 1$ is composite.

Exercise 2. With $y = 1$: $n^4 + 4 = (n^2 + 2n + 2)(n^2 - 2n + 2)$. The smaller factor is $n^2 - 2n + 2 = (n-1)^2 + 1 \geq 2$ for $n \geq 2$, and it is less than $n^4 + 4$, so the factorization is nontrivial.

Exercise 3. The step fails at $n = 1$: with two horses, removing one leaves a single horse, and the two singleton sets do not overlap, so there is no horse in common to transfer the color through. The step silently needs $n \geq 2$, so the basis would have to cover $n = 2$ as well, and of course $P(2)$ is false.

Exercise 4: Infinitely many primes of the form $4k - 1$

See the guide, page 11, Example 1.3(b). The skeleton:

Claim. There are infinitely many primes congruent to 3 (mod 4).

Proof. Suppose not, and let $p_1 = 3,\ p_2 = 7,\ \ldots,\ p_n$ be the complete list. Set $d = 4p_1p_2\cdots p_n - 1$.

- $d > 1$, so d has at least one prime factor.
- d is odd, so $2 \nmid d$.
- $d \equiv -1 \pmod{p_i}$ for every i, so no p_i divides d.
- Hence every prime factor of d is $\equiv 1 \pmod 4$: it is odd, and it is not $\equiv 3 \pmod 4$ because the p_i were all such primes.
- A product of integers $\equiv 1 \pmod 4$ is $\equiv 1 \pmod 4$, so $d \equiv 1 \pmod 4$.
- But $d = 4(\cdots) - 1 \equiv 3 \pmod 4$. Contradiction. □

Grading notes. The bullets "d is odd" and "a product of 1 mod 4 numbers is 1 mod 4" must both appear explicitly. Skipping either costs points.

Why it fails for $4k + 1$. A product of numbers $\equiv 3 \pmod 4$ can be $\equiv 1 \pmod 4$ (e.g. $3 \cdot 7 = 21$), so from "$d \equiv 1 \pmod 4$" you cannot conclude d has a prime factor $\equiv 1 \pmod 4$. The $4k + 1$ case needs a different idea: any odd prime dividing $m^2 + 1$ is $\equiv 1 \pmod 4$.